

R.O.B. VISION

VIRTUAL ROBOT COMPANION / MK.01



USER GUIDE

Play, preview, setup, and problem solving

CURRENT REFERENCE / 25 SEPTEMBER 2026
UNO Q + RETROPIE LIVE · FRAME LINK VERIFIED

CHAPTER 01 / CURRENT REFERENCE

Meet Buddy

Buddy is the little virtual companion at the heart of R.O.B. Vision. In our story, he has spent years inside an emulator, hearing instructions as flashes of light but never seeing who is on the other side. The game-frame link gives those signals a new destination. His first greeting is a blink on the UNO Q; his next is a movement on your screen.



Buddy is curious, patient, and happiest when you give him something to do. Watch his eyes, hands, and game pieces as each received command becomes a visible action. When the game ends, he waits for the next round.

This illustration is Buddy's story portrait. Mission shows the working virtual model and its exact game state; use the on-screen model and status indicators when following setup or gameplay steps. Buddy is an original character concept for this independent maker project. Nintendo's games and historical robot remain their owners' work.

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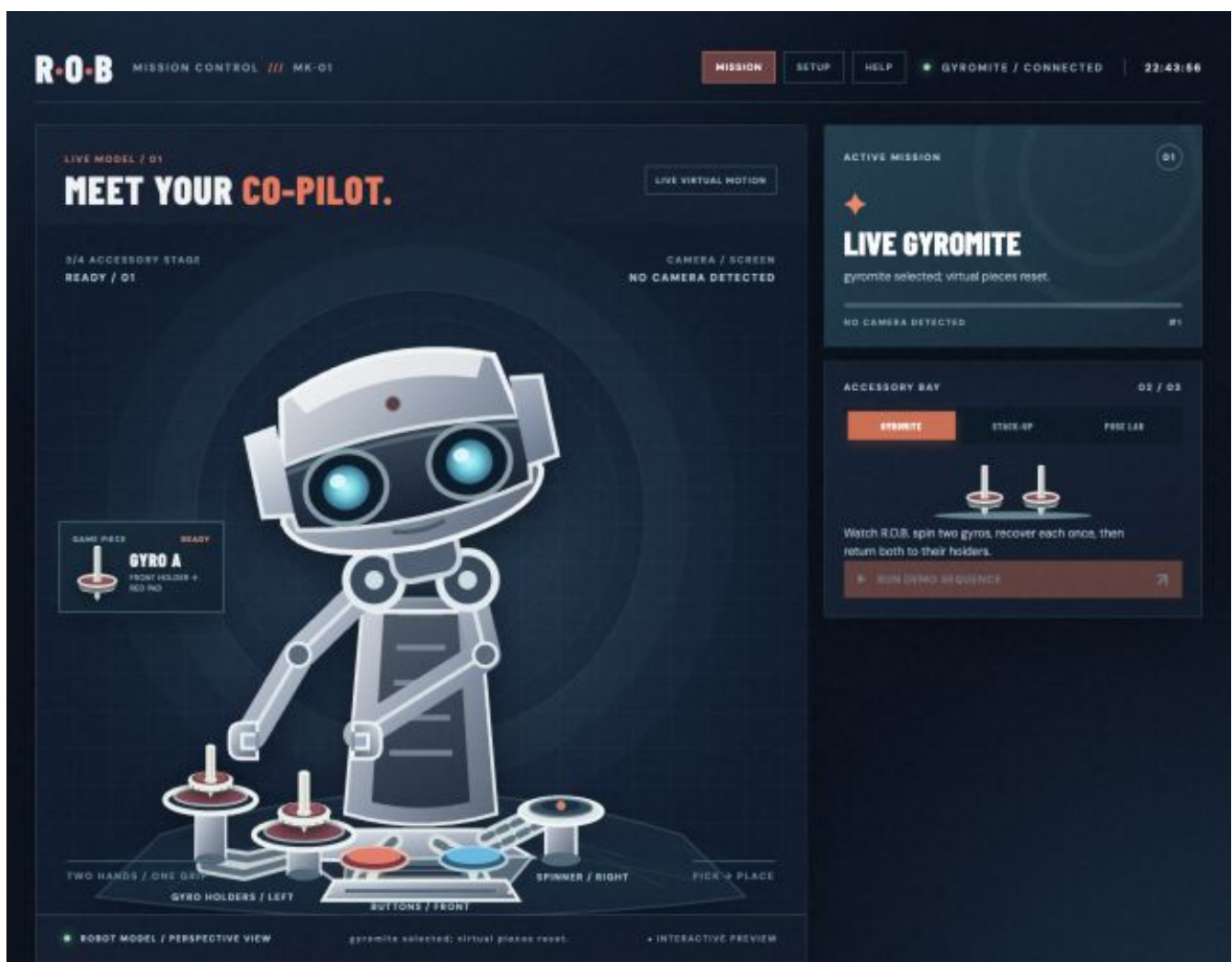
R.O.B. Vision user guide

The UNO Q hosts Buddy, our virtual robot companion, and receives Gyromite and Stack-Up commands through RetroPie's game-frame link. The robot, gyros, blocks, trays, and spinner are graphics and software state. There is no physical robot or camera. [Meet Buddy](#) introduces his story; the instructions below describe the working controls.

Open R.O.B. Vision

Start **R.O.B. Vision** in UNO Q App Lab, then open `http://arduino.local/dashboard/` on a laptop, phone, or tablet. Stop VirtualGlove first if it is running; this UNO Q runs one App Lab app at a time. The direct `http://arduino.local:8766/dashboard/` address reaches the same controller. A `file://` page is an offline preview.

Mission shows R.O.B., Accessory Bay, Game Table, Pose Preview, System Vitals, and Activity Feed. **GYROMITE / CONNECTED** means the browser reaches the UNO Q with Gyromite selected. **RETROPIE ONLINE** means the receiver has contacted the UNO Q recently. **GAME FRAMES LINKED** means the selected game is sending its rendered frames.



Try a demonstration

When no supported game is running, select **Gyromite** or **Stack-Up** and **Run Demo Sequence**. Gyromite demonstrates a held unspun gyro, two spinning gyros on pads, spin-down, re-spin, and return to holders. The illustrative spin clock is 55 seconds. Stack-Up moves red alone and then a blue/white group. **Home** resets the pieces. Pose Lab is a character-only movement sandbox. A live game launch stops the demo and selects that game's scene.

Choose an NES emulator

For automatic R.O.B. movement, select **lr-robvision-fceumm** or **lr-robvision-nestopia** for the game in RetroPie. Each entry uses the corresponding installed original core and forwards the game's rendered light cells to the UNO Q. The plain **lr-fceumm** and **lr-nestopia** entries play normally without the R.O.B. Vision frame link. Gyromite and Stack-Up currently default to the FCEUmm choice on the test RetroPie. See the [RetroPie setup guide](#).

Play Gyromite

Open [Setup](#) to check RetroPie and game-frame status. Start Gyromite and watch Mission for **GAME FRAMES LINKED**. R.O.B. follows complete commands from the game. A spinning virtual gyro on a pad keeps its gate pressed while R.O.B. moves elsewhere; an unspun gyro can press one gate while held there. The UNO Q sends those pad states to RetroPie's virtual Controller 2.

Fast Gates responds immediately while Gyromite is selected. Tap **Lower Blue** or **Lower Red** to press, tap again to release, or choose **Release Both**. Keyboard shortcuts are **2**, **1**, and **0**. Holds expire after 60 seconds; both buttons release on game exit or lost receiver data. If regular controls moved a gyro, use **Home** to restore its holder before Fast Gates. Both gate colors have been verified in Game A under FCEUmm. The game screen remains the authority for Hector's position and gate animation.

Play Stack-Up

The live model begins with five colored blocks on Tray 3. In Direct mode, jumping Hector onto a command tile sends **Left**, **Right**, **Up**, **Down**, **Open**, or **Close** through the frame link. Manual controls use the same virtual model. A lower grip carries the contacted block and all blocks above it. Impossible moves are rejected and leave pieces in place. **Home** restores the initial stack. The model does not score the game or initialize Bingo's distinct historical layout. Stack-Up has no Gyromite-style Controller 2 gate return.

Test mode and reset

Launch the game through its R.O.B. Vision emulator choice and enter Test mode. The game-frame link makes R.O.B.'s red light blink automatically on Mission and Setup when it recognizes the Test signal. A separate ready-light command makes the light steady. Neither moves R.O.B. **Preview Red Light** on Setup demonstrates the animation without claiming a game signal.

Home resets the selected virtual game. Live **Emergency Stop** clears the game selection; in the offline preview it cancels the script until reset. A browser reload restores the UNO Q's current snapshot. See [Setup](#), [gameplay](#), and [troubleshooting](#).

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R.O.B. Vision gameplay guide

Status: scripted preview and connected UNO Q controller. R.O.B., the gyros, spinner, pads, trays, and blocks exist only in the virtual model. The UNO Q controller owns their state and updates the browser. Red and blue pad presses reached RetroPie's virtual joystick, and the player confirmed both gate colors respond in Game A. The RetroPie frame link decoded all six command types in both games during live Direct play.

Gyromite

The original Gyromite booklet describes **Test**, **Direct**, **Game A**, and **Game B**. Direct sends six R.O.B. actions: up, down, left, right, open, and close. In Game A, Start switches into the robot-transmission screen for a command and then back to the professor. In Game B, the professor walks while Controller 1 directly commands R.O.B. A gyro on the blue or red tray lowers the matching gate.

The manual explicitly says a gyro **does not need to spin to operate only one gate**. R.O.B. can hold an unspun gyro on the virtual pad, then lift it to release that pad. If both gates must be controlled, spinning one gyro lets it stay upright on one pad while R.O.B. uses his hands to put the other on the other pad. A gyro that is no longer needed belongs on its holder. The spinner is a means to free R.O.B.'s hands, not an obligatory step for every pad press.

Available preview: the finite Gyromite demo shows an unspun held press, a two-gyro spin relay, separate spin-down and one recovery for each gyro, then a tidy return to both holders. It stops at COMPLETE. The spin lifetime is illustrative. Select **Home** to reset. In the connected setup, UNO Q pad state reaches RetroPie's virtual Controller 2; the browser's local-only preview does not drive that receiver.

Live controller: the UNO Q maintains the locations and spin phases of Gyro A and B. It derives red and blue virtual button states from those modeled conditions and sends them over LAN to the paired RetroPie Controller 2 receiver. It releases both buttons on game exit, connection loss, stale packets, or mode change. The player confirmed that both red and blue gate controls work in Game A with the corrected Controller 2 mapping. The **Fast Gates** dashboard controls place or return the matching virtual gyro and send the pad state immediately, with a 60-second automatic release. This keeps gate operation responsive while the player moves Hector. The RetroPie frame link controls R.O.B.'s virtual actions from complete rendered-frame commands; the launch hook supplies the game ID. The controller keeps game context and modeled piece state distinct from unknown on-screen professor or gate positions.



Pose Preview, Fast Gates, and the simulated Game Table on Mission.

Stack-Up

The [Stack-Up booklet](#) describes Direct, Memory, and Bingo. The virtual fixture has five numbered trays and five colored blocks. Direct and Memory start with all five on Tray 3, top to bottom **red, white, blue, yellow, green**. One-player Bingo historically uses any color order on Tray 3; two-player Bingo starts with three on Tray 3 and one each on Trays 2 and 4. The current live model initializes the Direct/Memory stack for all modes; Bingo-specific layouts are not implemented. Direct moves Professor Hector onto six command keys: **left, right, up, down, open, close**. Memory plays a programmed command series; Bingo sends a command when one row or column completes. The UNO Q code applies each validated optical command as one virtual action; all six command types were observed through the live RetroPie frame link. It must model the grippers' height, so R.O.B. can carry a block together with all blocks above it. Stack-Up's documented modes do not need Gyromite's virtual Controller 2 pad return path, even though two-player Bingo uses a second human controller.

Available preview: R.O.B. first moves red alone from Tray 3 to Tray 4, then grips blue and white together and carries both to Tray 2. Green and yellow remain on Tray 3. All five blocks appear in the Live Model, Accessory Bay, and Game Table as they move. The **Stack-Up Commands** buttons above System Vitals apply one virtual left, right, up, down, open, or close action. **Home** restores the starting stack. A blocked move leaves all blocks in place and explains why. The preview demo is scripted; the separate live controller accepts manual commands and complete ROM-pattern detections from RetroPie game frames. In game play, the player compares the virtual arrangement with the target shown by the game. The flash stream does not tell the controller which block to move or whether the target was achieved.

Pose Lab and recovery

Pose Lab exercises R.O.B.'s head, body turn, arms, and paired grip without game pieces or commands. Its demo is a short greeting; the controls below the scene let the viewer try poses manually. It is an art and motion sandbox, not a third game. In a live session, unreadable flashes produce no movement. A browser reconnect receives the UNO Q's current snapshot; it never restarts an action. If the Gyromite return path disconnects, virtual Controller 2 buttons release even if the browser still shows a gyro on a pad. The game host receives controller states, not a proof that a level was completed.

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R.O.B. Vision Setup

Open `http://arduiain.local/dashboard/setup.html`. Console Link lists every saved RetroPie and Batocera console separately, with its address and live status. Setup has four checks:

On Batocera 43.1 x86_64, first install with `python3 scripts/install_batocera.py --controller arduiain.local` from a staged checkout. For first pairing run `python3 /userdata/system/rob-vision/tools/retropie_pair.py --platform batocera` and enter its code and fingerprint on Setup. The exact Gyromite and Stack-Up ROMs automatically use the R.O.B. Vision FCEUmm wrapper; Nestopia is an optional per-ROM choice. [Batocera installation details](#).

1. **Pair the console.** Install the receiver and launch hooks first. On RetroPie run `python3 /home/pi/rob-vision/tools/retropie_pair.py`. For five minutes it shows a six-digit code and SHA-256 fingerprint. Enter its hostname, code, and fingerprint on Setup, then select **Pair Console**. The UNO Q verifies the certificate fingerprint before sending a unique console credential over the encrypted pairing connection. Repeat for another console using **Pair Another Console**. Each appears under **Paired Consoles**; **Check Link** counts recent authenticated polls, and an individual row goes offline after three seconds without one. Pairing does not install the receiver or emulator choices.
2. **Check game frames.** On RetroPie, select `lr-robvision-fceumm` or `lr-robvision-nestopia`. On Batocera, launch an exact registered ROM; its R.O.B. Vision FCEUmm wrapper is selected automatically. Setup should show **GAME FRAMES LINKED**. The wrapper forwards each rendered NES light cell to the local receiver, which sends complete commands to the UNO Q. The plain `lr-fceumm` and `lr-nestopia` entries play without this link.
3. **Check Test mode.** Launch the game through its R.O.B. Vision emulator choice and enter Test mode. Gyromite's sustained green field or Stack-Up's alternating green/white frames make R.O.B.'s red head light blink automatically on Mission and Setup. A separately decoded ready-light command makes it glow steadily. Neither moves arms or game pieces. **Preview Red Light** only demonstrates the animation. If the light stays off, check **GAME FRAMES LINKED** and confirm the game is in Test mode. In Stack-Up, press Select at the ROBOT BLOCK artwork to show the mode list, choose Test, then press Start.
4. **Try the controls.** For Gyromite, **Lower Red** and **Lower Blue** press matching virtual pads immediately; press again to release or choose **Release Both**. Each hold expires after 60 seconds. For Stack-Up, **Left, Right, Up, Down, Open, Close** send one command to the virtual model. Stack-Up has no Gyromite-style Controller 2 return path. Watch Mission's Live Model and Game Table for movement.

A `file://` Setup page is only an offline preview. Use the UNO Q URL for live status. Mission keeps Fast Gates and the Game Table next to Pose Preview and System Vitals.

To forget a console, select **Remove** in its row and then **Confirm Remove**. The UNO Q immediately revokes that console's credential; any other paired console remains connected. If the removed console was playing, its game session is cleared. The removed console retains its old local credential, but it can no longer authenticate; run its pairing command again to reconnect it. Only one console can be the active game source at a time.

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R.O.B. Vision installation guide

Installed test setup, 25 September 2026: R.O.B. Vision is a separate UNO Q App Lab app. `retropie.local` has exact-name launch/exit hooks, pairing, a virtual Controller 2 receiver, and FCEUmm and Nestopia frame-link choices. Gyromite launch/exit and both gate colors were verified under FCEUmm. Both games delivered commands through FCEUmm and Nestopia. Nestopia's Gyromite Player 2 gate return still needs a Game A check. There is no physical robot or camera. The installer below is a development preview. It passed local tests and live in-place upgrades on both devices, but a fresh installation on two new devices has not yet been run.

Before you begin

- Use an Arduino UNO Q with App Lab and either standard RetroPie with the `pi` account and `/opt/retropie/configs`, or Batocera 43.1 x86_64. The UNO Q needs Python 3.9 or newer; RetroPie needs Python 3.7 or newer. Direct cloning needs `git` on each device; the SSH archive transfer below does not. RetroPie also needs `gcc`, `openssl`, `modprobe`, `systemd`, and at least one installed NES libretro core: `lr-fceumm` or `lr-nestopia`.
- Supply your own legally obtained Gyromite or Stack-Up ROM with an exact filename listed in `config/games.json`. The repository and installer contain no ROMs.
- Keep both devices on a trusted local network. Exit any NES game before installation or upgrade. The UNO Q installer stops and restarts R.O.B. Vision automatically; stop any other App Lab app first. The UNO Q can run only one App Lab app at a time on the tested device.
- The GitHub repository is currently private. A direct clone on a device requires a GitHub account with access and authenticated Git on that device. You can instead send the source from an authorized computer over SSH without copying GitHub credentials to either device.

For the SSH transfer option, run these commands **on a computer with a local checkout of dev**, replacing the device names. They create only the source folders used by the installer:

```
git archive --format=tar dev | ssh arduino@your-uno-q.local 'mkdir -p /home/arduino/rob-vision-src && tar -xf \
- -C /home/arduino/rob-vision-src'
git archive --format=tar dev | ssh pi@your-retropie.local 'mkdir -p /home/pi/rob-vision-src && tar -xf - -C /
/home/pi/rob-vision-src'
```

Then skip each `git clone` command below and run its installer command. For an upgrade using this option, transfer the new archive again before rerunning the installer.

Install on the UNO Q

As the `arduino` user, clone the development branch outside the App Lab app folder and run:

```
git clone --branch dev https://github.com/mathan416/ROB-Vision.git ~/rob-vision-src
python3 ~/rob-vision-src/scripts/install.py uno-q
```

The installer stages the app in `~/ArduinoApps/rob-vision`, including its `sketch/sketch.ino` matrix-display program. It creates a private controller token on first installation and preserves the token and other app data on upgrade. It stops the running R.O.B. Vision app, installs the new version, starts it through App Lab, and waits for the controller to respond. App Lab compiles and uploads the display sketch when it starts the app; this does not replace the UNO Q's board firmware or bootloader. The installer retains the prior app in a `rob-vision.previous*` sibling for rollback. If the new app fails to start, it restores and restarts the previous version. The normal panel address is `http://<your-uno-q-hostname>.local/dashboard/`; open `/dashboard/setup.html` to pair the

console. App Lab owns ports 80 and 8766 while the app runs. Do not enable the separate `deploy/rob-vision.service` at the same time.

On upgrade, the installer also preserves App Lab's `.deps` and `.cache` folders. The previous app directory remains available for rollback until you remove it.

Install on RetroPie

As `pi`, clone the same branch and run the installer with the UNO Q's LAN hostname or IP:

```
git clone --branch dev https://github.com/mathan416/ROB-Vision.git ~/rob-vision-src
sudo python3 ~/rob-vision-src/scripts/install.py retropie --controller your-uno-q.local
```

It installs the root virtual Controller 2 receiver, builds frame wrappers for installed FCEUmm and/or Nestopia cores, and adds two NES launch choices. It selects those choices only for the registered Gyromite and Stack-Up filenames, preserving any unrelated custom emulator choice. Existing `runcommand` scripts stay in place; the installer adds a marked R.O.B. Vision section near the top and writes a one-time `.before-rob-vision` backup. Its NES config edit is also marked and backed up. It stores the controller address in `/home/pi/.config/rob-vision/receiver.env` and loads `uinput` on future boots. Your ROMs are not copied or changed.

Pair the receiver before starting its service:

```
python3 /home/pi/rob-vision/tools/retropie_pair.py
```

Enter the six-digit code and SHA-256 certificate fingerprint in the UNO Q Setup page, then run:

```
sudo systemctl enable --now rob-vision-controller2.service
sudo python3 ~/rob-vision-src/scripts/install.py player2
```

The last command discovers **R.O.B. Vision Controller 2** among RetroPie's joysticks and writes its actual index plus the tested red/blue mapping to NES RetroArch configuration. It leaves Player 1 alone. Exit and relaunch Gyromite so RetroArch reads the new mapping. If a joystick order changes later, rerun `player2` while the receiver is running. If you know the index in advance, `retropie --player2-index N` sets it during the initial installation.

In RetroPie's per-game emulator selection, choose `lr-robvision-fceumm` or `lr-robvision-nestopia`. On Setup, **Check Link** should show the receiver online, and launching a registered ROM through a R.O.B. Vision choice should show **GAME FRAMES LINKED**. Start with Gyromite Test, then test blue and red gates individually in Game A. Stack-Up Direct mode can verify each of its six movements. The installer does not reprogram an existing Player 1 controller or its EmulationStation mapping.

To upgrade, exit the games, update each `~/rob-vision-src` checkout, and rerun the corresponding installer. The UNO Q installer handles the App Lab restart. Existing paired consoles are preserved. The console installer restarts the receiver when its credential already exists; no fresh pairing is needed unless that console was removed from Setup or its key changed.

Live installer check: On `retropie.local` (Python 3.7), the installer migrated the original standalone hooks, rebuilt both wrappers, preserved the paired token, restarted the receiver, and detected virtual joystick 1. A repeat installation left one managed hook per event and one Controller 2 mapping. Installed launch/end hooks selected and cleared Gyromite and Stack-Up on the UNO Q. The controller reported an authenticated online receiver after the restart. The exact published `dev` archive was transferred over SSH and installed successfully. A direct unauthenticated GitHub clone failed because the repository is private; the check did not run a new gameplay session or exercise first-time pairing on a blank RetroPie.

UNO Q installer check: On `arduian.local` (Python 3.13), an in-place upgrade automatically stopped the running R.O.B. Vision app through App Lab, installed the update, compiled and uploaded the sketch, and restarted the app. Mission, Setup, Help, and controller endpoints returned HTTP 200; `/api/matrix/state` reported an available, connected bridge in idle mode. The pairing token matched the previous app, and RetroPie remained paired and online. Earlier upgrades confirmed that `.deps` and `.cache` survive. Rollback on a failed App Lab start passed an installer test; it has not been forced on the live UNO Q. A blank-device first install remains untested.

Install on Batocera

On Batocera 43.1 x86_64, copy this checkout to a persistent path outside `/userdata/system/rob-vision`, then run as root while no game is open:

```
python3 scripts/install_batocera.py --controller your-uno-q.local
```

The installer enables a separate `ROBvision` service and launch hook, adds bundled x86_64 FCEUmm and Nestopia frame wrappers, and selects FCEUmm for the exact registered ROM filenames. It leaves VirtualGlove's service and Super Glove Ball choice intact. To pair for the first time, run `python3 /userdata/system/rob-vision/tools/retropie_pair.py --platform batocera`, then enter its code and fingerprint on Setup. The service detects the virtual pad and saves **Gyromite-only** RetroArch Player 2 overrides in Batocera's persistent per-ROM settings. Batocera regenerates its live RetroArch config on every launch, so editing `retroarchcustom.cfg` by hand will not persist.

The test Batocera installed and repeated the bundled installer, loaded both wrappers for each game through real launches, selected Stack-Up's FCEUmm wrapper without an explicit core override, retained Gyromite's Player 2 index and A/B mapping after config generation, and showed **Batocera online / game frames linked** on the UNO Q during Stack-Up. A live Batocera gate-response test and full playthrough are still open. See the [Batocera deployment guide](#) for core choice and architecture limits.

UNO Q and browser

Start **R.O.B. Vision** under App Lab **My Apps**. Stop VirtualGlove first because the tested UNO Q runs one App Lab app at a time. App Lab exposes port 80 at `http://<your-uno-q-hostname>.local/dashboard/`; the controller is also reachable on port 8766. Both addresses share one virtual game state. The App Lab `python/main.py` gateway talks to `controller/service.py`, and Router Bridge drives the matrix sketch. Avahi supplies the `.local` hostname when configured on the UNO Q. Keep the separate `deploy/rob-vision.service` disabled while App Lab owns port 8766.

Open [Setup](#) to check pairing and the game-frame link. Browser controls are available on the trusted LAN without a token prompt. New pairings use a separate credential for each console, stored privately in `data/paired-consoles.json` on the UNO Q. The installer's `data/controller-token` remains a migration credential for older paired consoles; an existing `ROB_VISION_TOKEN` environment setting overrides that file. Console Link shows each saved console and can remove it independently. Keep the LAN private. A local `file://` page is an offline preview.

RetroPie

Follow the [RetroPie deployment instructions](#) for the launch/end hooks, root uinput receiver service, pairing, FCEUmm and Nestopia launch choices, and RetroArch port 2 configuration. Merge hooks with any existing scripts rather than replacing unrelated commands. After pairing, **Check Link** turns online when authenticated receiver polls arrive. The receiver can resync game identity from the active RetroArch process after a UNO Q restart.

For Gyromite or Stack-Up, choose `lr-robvision-fceumm` or `lr-robvision-nestopia` in RetroPie's emulator selection. These entries run the installed original core through R.O.B. Vision's frame wrapper. Plain `lr-fceumm` and `lr-nestopia` play without automatic R.O.B. movement. The UNO Q receives only complete game-specific light commands. The Test light responds automatically through the same frame link.

Local development

Run `python3 -m controller.service` and open `http://127.0.0.1:8766/dashboard/`. For trusted-LAN access set `ROB_VISION_TOKEN` and use `--host 0.0.0.0`; the token protects RetroPie traffic, while browser controls are available on that LAN. The static preview can be served with `python3 -m http.server 8000` and has no game link.

Run the [verification plan](#) before a general release.

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R.O.B. Vision troubleshooting

First check which page is open: `http://arduiain.local/dashboard/` is live; a `file://` page or a simple static web server shows the independent preview.

Symptom	Check
App does not start, port 8766 in use	Stop VirtualGlove or the separate <code>rob-vision.service</code> ; App Lab and that service cannot both bind the port.
GYROMITE / CONNECTED but no robot movement	Check RETROPIE ONLINE and GAME FRAMES LINKED separately. If the latter is absent, restart the game so RetroPie loads the R.O.B. Vision core. Check Mission's Activity Feed for decoded or blocked actions.
RetroPie is online but no game is selected	A receiver can be online with no active game. Launch a registered ROM; then reload Mission if needed.
Test light never blinks	Enter the game's Test mode and confirm GAME FRAMES LINKED with the correct game selected. The light responds automatically to linked game frames. A red-light preview is only an animation demonstration.
Movement command is missed	On RetroPie, confirm GAME FRAMES LINKED and the correct game is selected. The frame link reads every rendered NES frame; check the Activity Feed for a decoded command or a model move blocked by height, grip, or station.
Red/blue gate does not move	Confirm Gyromite is the active RetroArch game, Setup Check Link is online, and RetroArch port 2 is configured. Inspect the game screen during an isolated color hold.
Fast Gates vanished	They show only in live Gyromite.
Gate remains down	Select Release Both ; a hold also expires at 60 seconds. The receiver releases on game exit or stale network data.
Stack-Up block does not move	Review current station, height, gripper, carried group, and destination capacity; invalid moves leave the stack intact and appear in activity.
Demo does not start	Exit the running game first. Demo mode works when paired and idle; Home resets it.
Browser shows stale scene	Reload for a fresh <code>/api/state</code> snapshot. A local preview cannot show UNO Q state.

On the dedicated RetroPie test setup, `systemctl status rob-vision-controller2.service` checks the root virtual-controller service. Inspect RetroPie's `runcommand` log and configured ROM basename if launch identity is missing. `/dev/input/js1` is expected for the virtual Controller 2 while the player's controller is generally `/dev/input/js0`. Keep tokens and ROM files out of bug reports.